



The American University in Cairo

DSCI 2411 - Data Visualization

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Project Report

Analysis of the Global Happiness Dataset

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1. **Introduction**

Many countries around the world have very different levels of happiness and quality of life. Some countries report very high happiness scores, while others report lower scores. This raises an important problem: what factors may explain why people in some countries report being happier than people in others? This dataset can help explain this problem by allowing us to compare happiness scores with country-level conditions such as human development, income, corruption, democracy, safety, and social well-being. This is important because happiness isn't only affected by personal life. It can also be affected by the wider environment people live in, including economic stability, quality of government, public safety, education, healthcare, and general living standards.

Originally, the data was provided as separate datasets for different variables, such as happiness score, HDI, income per person, CPI, democracy score, population, murder, and suicide. To make the analysis easier and more organized, we combined all the separate datasets into one master dataset using Excel. This allowed us to place all the important variables together in one table, where each row represents a country in a specific year. Combining the datasets made it easier to clean the data, compare variables, create visualizations, and analyze relationships between happiness and other factors.

The main variable in this project is the Happiness Score, which shows how people in each country rate their overall quality of life. The dataset also includes other variables that may help explain differences in happiness. HDI, or Human Development Index, measures development through health, education, and living standards. Income per Person shows the economic level of a country. CPI, or Corruption Perception Index, measures perceived corruption, where higher CPI values mean lower perceived corruption. Democracy Score shows how democratic or authoritarian

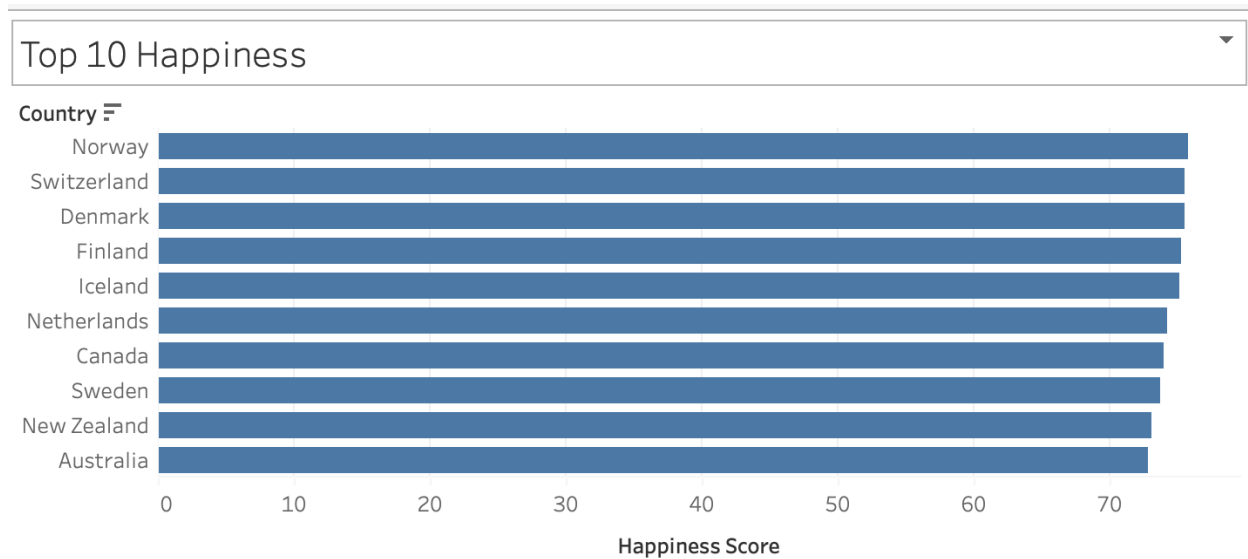
a country is. This dataset also includes Population, Murder Rate, and Suicide Rate, which help describe safety and social conditions.

The main hypothesis of this project is that countries with stronger governance and higher socioeconomic development tend to have higher happiness scores. This means that countries with higher HDI, higher income, higher CPI scores, stronger democracy, and lower instability indicators are expected to report higher happiness levels. The null hypothesis is that there is no clear relationship between happiness scores and these factors. In other words, if the null hypothesis is true, happiness wouldn't show a clear connection with development, income, corruption, democracy, or safety.

To test this hypothesis, the project will use data visualizations in three main parts. First, the project will summarize the dataset by showing the happiest and least happy countries, the distribution of happiness scores, the countries with the highest HDI, and the global map of happiness. Second, it will compare happiness with other variables such as HDI, income, CPI, and safety indicators to see whether clear relationships appear. Third, it will look at the changes over time to see whether countries that improve in development, income, or governance also show changes in happiness. The final goal is to use the visualizations and their insights to decide whether the data supports the main hypothesis or whether they fail to reject the null hypothesis.

2. Objective 1: Data Summarization

a) Top 10 Countries by Happiness Score (2012-2017)



This visualization is a horizontal clustered bar chart showing the top 10 countries with the highest average happiness scores across 2012-2017. The x axis represents the happiness score, while the y axis represents the countries. The bars are sorted in descending order, making it easy to compare the countries using bar lengths.

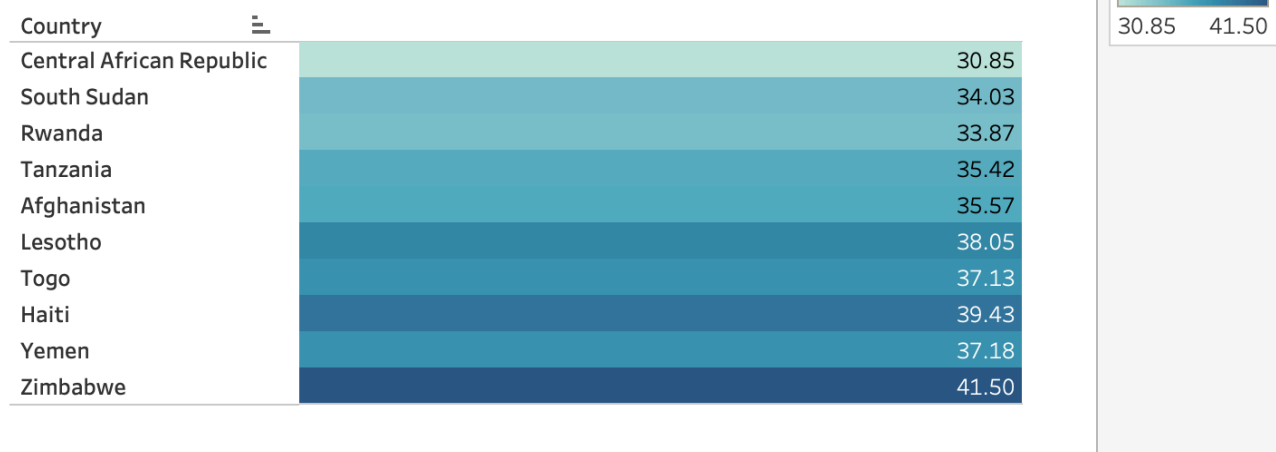
The strongest pattern is that the happiest countries aren't randomly distributed. Most of them are high income, highly developed, and politically stable countries, especially from Northern and Western Europe. This is important because these countries usually perform well in the same areas used in our hypothesis: human development, income, low corruption, democracy, and safety. This chart also suggests that happiness' connected to more than just economic wealth. If happiness depended only on income, we would expect the ranking to include only the richest countries. However, the countries shown here are also known for strong public services, reliable healthcare, good education, and social support systems. This supports the idea that national happiness' shaped by both economic and non economic factors. Another important point is that the bars are very close

in length. Norway has the highest average score, but the gap between the top 10 countries isn't very large. In other words, the top group forms a cluster rather than one extreme outlier. This matters because it suggests that high happiness may be linked to a shared set of national conditions, not just one unique country case.

The top happiest countries are mostly developed, stable, and high income countries. This gives early support to the hypothesis that stronger development and governance are linked to higher national happiness.

b) Bottom 10 countries by Happiness Score (2012-2017)

Bottom 10 Countries by Happiness Score



This heatmap shows the bottom 10 countries with the lowest average happiness score across 2012-2017. The countries include the Central African Republic, South Sudan, Rwanda, Tanzania, Afghanistan, Lesotho, Togo, Haiti, Yemen, and Zimbabwe. The color scale represents the average happiness score, where lighter shades show lower values and darker shades show high values within this bottom group.

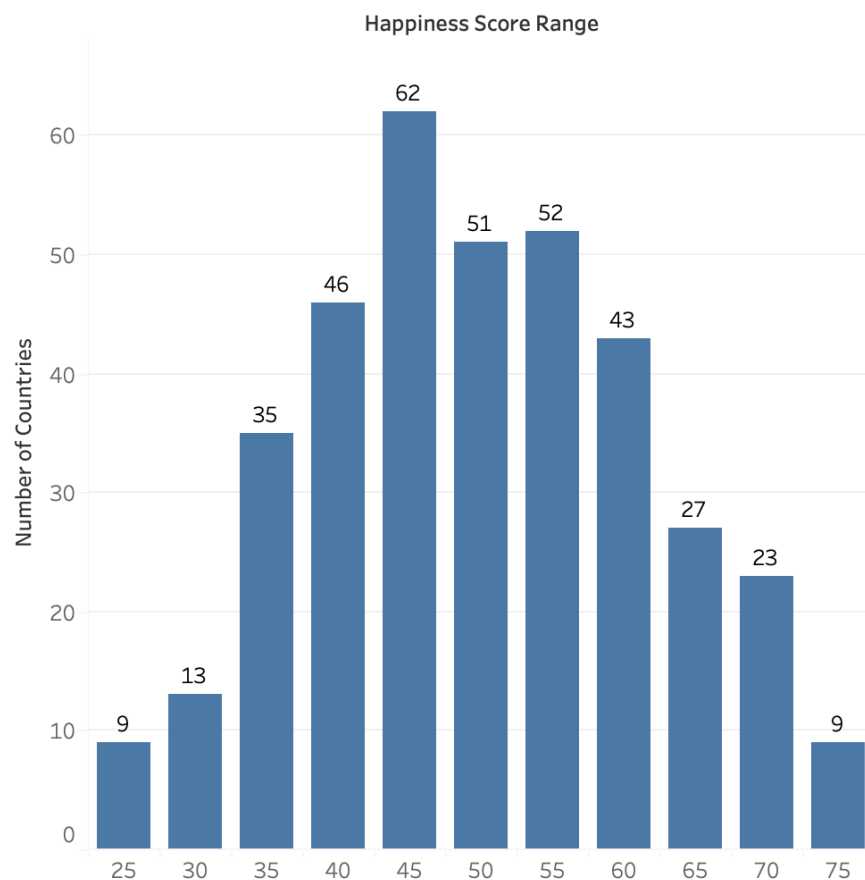
This visualization shows the opposite side of the first chart. While the top 10 happiest countries were mostly developed, stable, and high income countries, the bottom 10 countries are

mostly facing harder national conditions. Many of these countries have experienced poverty, conflict, weak institutions, political instability, limited public services, or safety challenges. These conditions can affect people's quality of life and reduce overall happiness. The heatmap's useful because it doesn't only rank the countries, it also shows the intensity of the values using color. The Central African Republic has the lowest average happiness score at 30.85, while Zimbabwe has the highest value within the bottom group of 41.50. This means that even among the lowest ranked countries, there's still a noticeable difference in happiness levels. Compared with the top 10 chart, the gap is large. The happiest countries scored in the 70s, while the bottom countries are mostly between the low 30s and low 40s. This shows a major difference in national well being between countries. It also suggests that happiness' connected to the broader environment people live in, including development, governance, income, safety and social stability.

This heatmap strengthens the pattern from the first visualization. The highest happiness countries were mostly developed and stable, while the lowest happiness countries are more connected with weaker development and instability. Together, both charts show that happiness isn't randomly distributed, it appears to be strongly connected to country level living conditions.

c) Happiness Distribution (2012-2017)

Happiness Distribution



This histogram shows the distribution of average happiness scores across countries. The x axis represents happiness score ranges, while the y axis shows the number of countries that fall into each range. The tallest bar is around the 45 happiness score range, with 63 countries, meaning this is the most common range in the dataset.

This visualization gives a wider view of the dataset because it doesn't only focus on the top or bottom countries. Instead, it shows how happiness scores are spread across all countries. The main pattern is that countries are concentrated in the middle happiness range, especially between 40 and 60. This means that most countries aren't extremely happy or extremely unhappy, they fall

somewhere in the middle. This distribution also shows that fewer countries are found at the two extremes. There are only a small number of countries in the lowest ranges around 25-30, and only a few countries reach the highest ranges around 70-75. This matches what we saw in the first 2 visuals, the happiest and least happy countries are special groups, while the majority of the countries are clustered around moderate happiness levels. Another important insight is that happiness' less common globally. The top 10 countries from the first chart are part of the smaller right side of the histogram, where fewer countries exist. This suggests that reaching very high happiness scores requires strong national conditions, such as better development, higher income, stronger institutions, and more stable social environments.

This histogram connects the first 2 visuals by showing the full distribution between the highest and lowest countries. Most countries fall in the middle range, while only a smaller group reaches very high happiness scores. This supports the idea that high national happiness isn't common everywhere and may be linked to stronger development, governance, and living conditions.

d) Top 10 HDI (2012017)



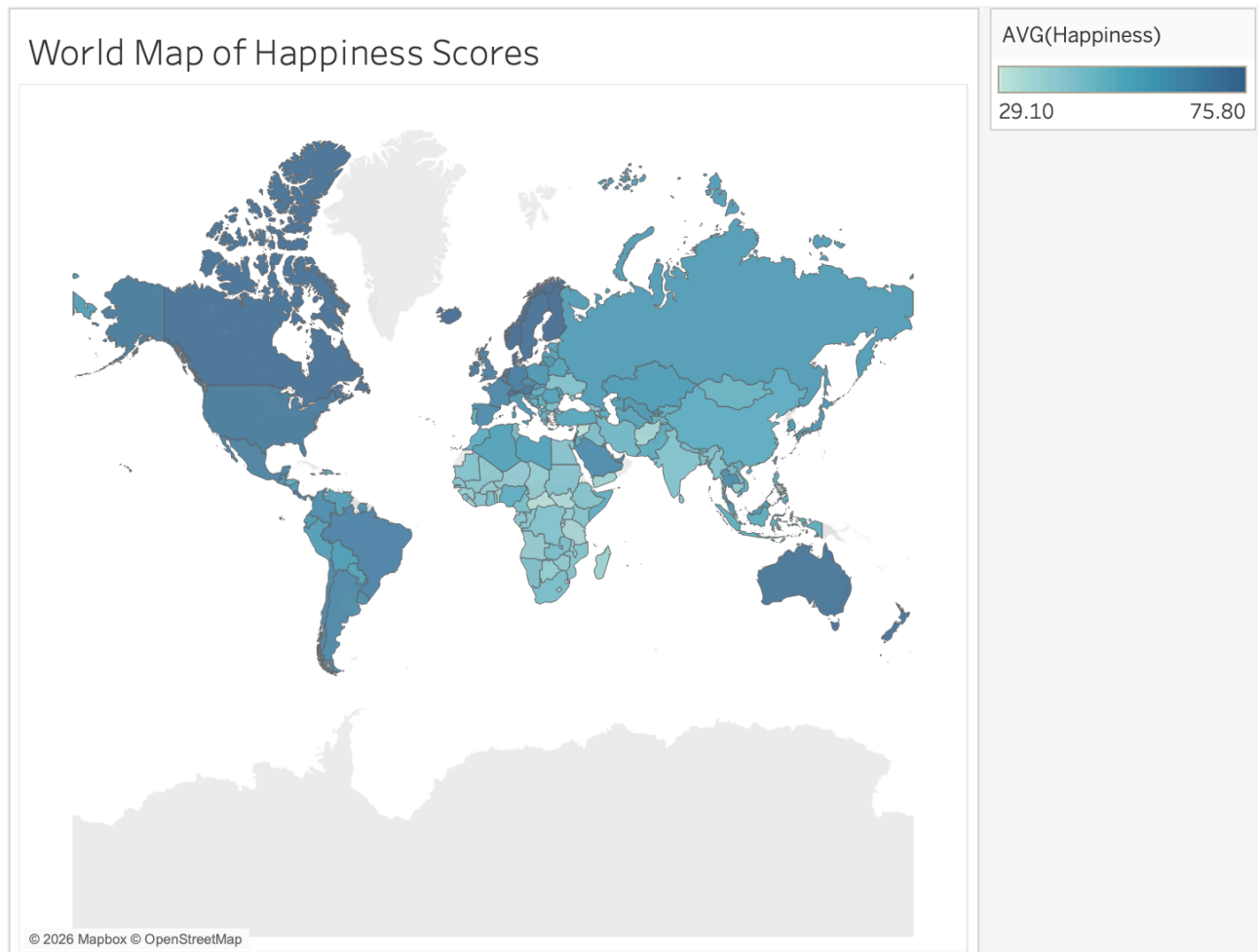
This packed bubble chart shows the top 10 countries with the highest average Human Development Index (HDI). Each bubble represents a country, while the color intensity represents the average HDI value. Countries shown include Switzerland, Norway, Germany, Iceland, Australia, Denmark, Ireland, Sweden, Finland, and the Netherlands.

This visualization focuses on human development, which is one of the main factors in our hypothesis. HDI is a strong indicator because it doesn't only measure income. It combines health, education, and living standards, so it gives a wider view of quality of life in each country. A clear pattern is that many of the countries with the highest HDI also appeared in the top 10 happiness chart. Countries such as Norway, Switzerland, Denmark, Finland, Iceland, the Netherlands,

Sweden, and Australia are present in both visuals. This overlap's important because it suggests that countries with stronger human development also tend to have higher happiness scores. The bubbles are close in color range, which means the top HDI countries have similar development levels. This shows that they form a high development cluster rather than one country being far ahead of the others. These countries likely share strong national conditions, such as better healthcare, stronger education systems, higher living standard, and more economic stability.

This chart adds more depth to the earlier happiness visuals. The top happiness countries and top HDI countries overlap strongly, showing that high happiness' often found in countries with strong human development. This supports the hypothesis that better health, education, and living standards are linked to higher national happiness.

e) World Map (2012-2017)



This filled map shows the average happiness scores across countries. The color scale represents the average happiness score, where lighter colors show lower happiness values and darker colors show higher happiness values. This type of visualization is useful because it shows geographic patterns that are harder to see in ranking charts or tables.

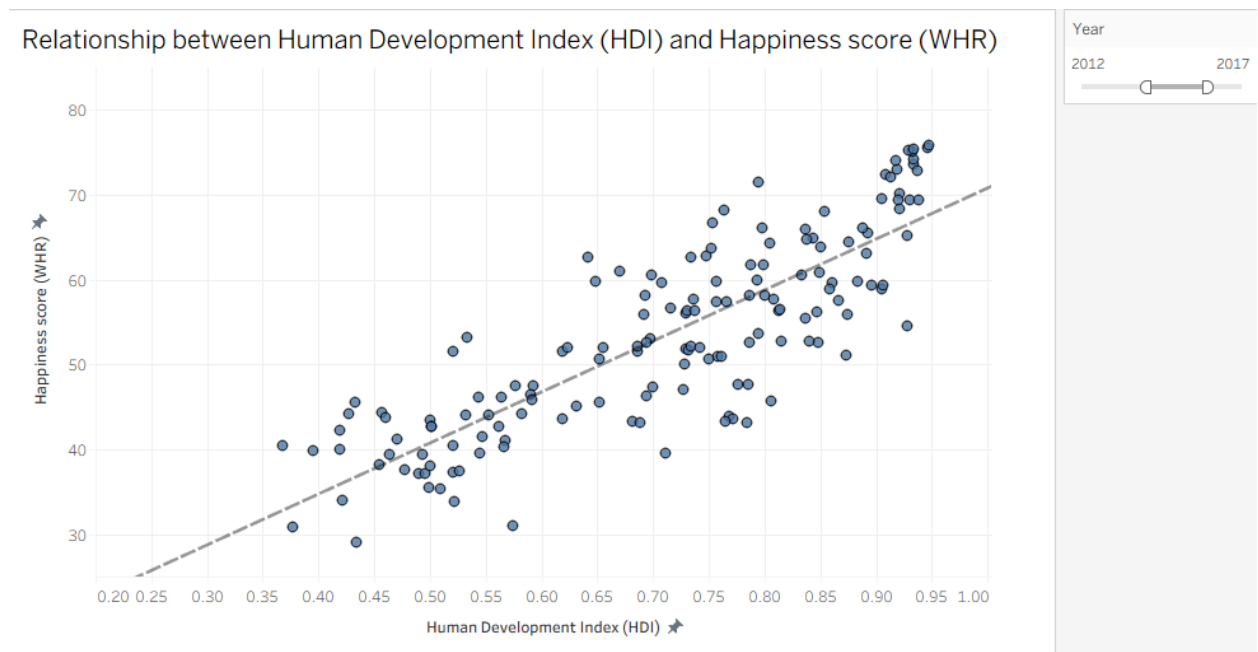
The map shows that happiness isn't evenly distributed across the world. Higher happiness scores are mainly concentrated in North America, Northern and Western Europe, and Oceania, especially countries such as Canada, Australia, New Zealand, and many European countries. These regions also appeared in the earlier top happiness and top HDI visuals, which makes the pattern more consistent. On the other hand, lower happiness scores appear more often in parts of Africa, the

Middle East, and some lower development or conflict affected regions. These lighter colored areas suggest that countries facing weaker development, political instability, safety problems, or lower living standards tend to report lower happiness scores. This map adds a spatial layer to the analysis. The previous visuals show the highest countries, lowest countries, and the overall distribution, but this map shows where those patterns are located geographically. The clustering of darker and lighter regions suggests that national happiness is influenced by broader regional and country level conditions, not just individual country cases.

This map connects the previous objective 1 visuals together. The countries and regions with higher happiness scores are generally developed and stable, while lower scores appear more often in regions facing economic, political , or safety challenges. Overall, the geographic pattern supports the hypothesis that happiness' linked to development, governance, income, and social stability.

3. Objective 2: Identifying Relationships between Attributes

a) Relationship Between HDI and Happiness Score



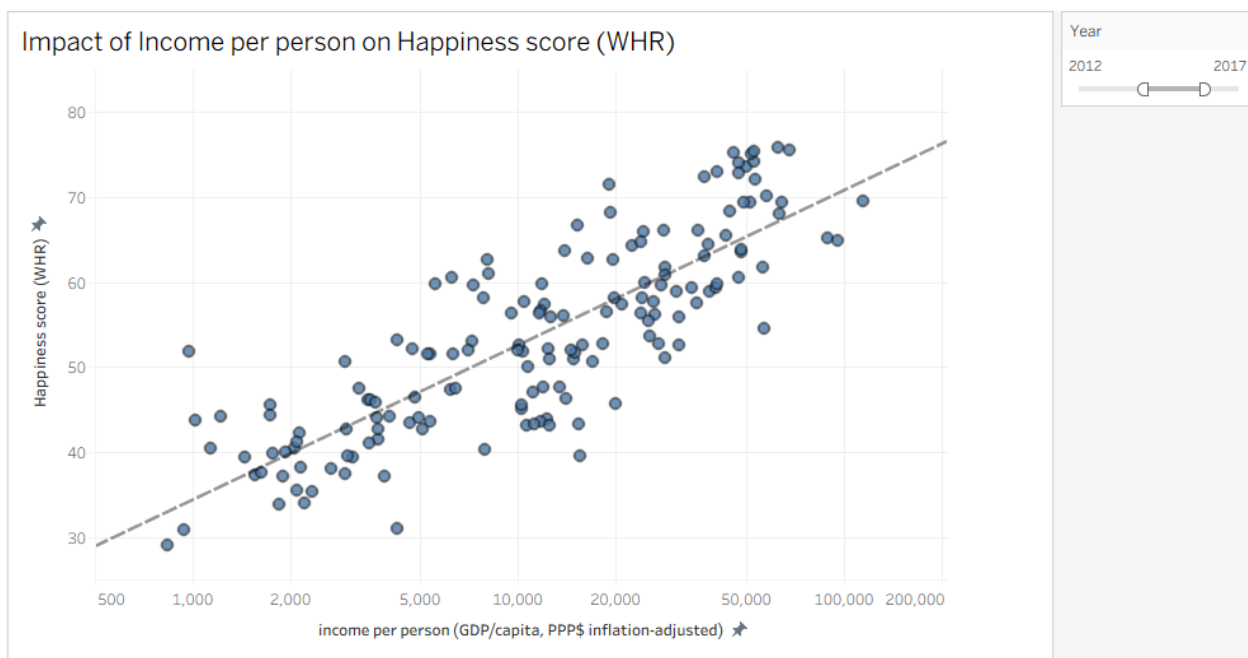
This scatterplot compares the Human Development Index (HDI) with the Happiness Score (WHR) for countries between 2012 and 2017.

The scatterplot displays a clear upward trend, indicating that countries with higher HDI values generally tend to report higher happiness scores. This relationship is reasonable because HDI reflects important aspects of quality of life such as health, education, and living standards. Countries that perform well in these areas are more likely to provide better conditions for their populations, which can contribute to higher levels of happiness. Most countries are concentrated around the trend line, which suggests a relatively consistent relationship between the two variables. Countries with low HDI values are mostly concentrated within the lower happiness range, while countries with high HDI values are concentrated in the upper happiness range. The graph also shows that the relationship becomes more stable at higher HDI levels. Countries with HDI values above approximately 0.80 show less variation compared to countries in the middle HDI range.

Although a few countries fall noticeably above or below the trend line, these outliers do not significantly affect the overall pattern. The R-squared value of 0.7027 indicates a strong relationship between HDI and the very small p-value (less than 0.0001) suggests that the relationship shown in the graph is unlikely to be caused by random variation

Overall, the visualization shows a strong positive relationship between HDI and happiness. Countries with higher levels of human development generally tend to report higher happiness scores.

b) Impact of Income on Happiness



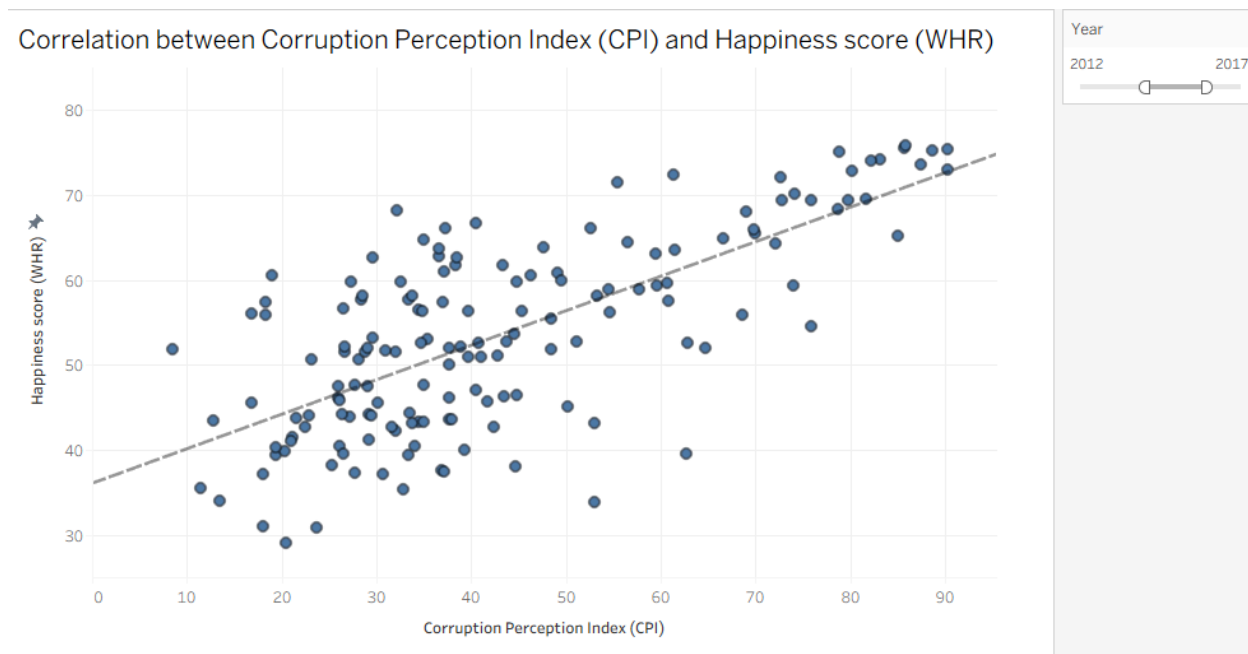
This scatterplot illustrates the relationship between Income per person and Happiness Score across countries between 2012 and 2017.

The scatterplot shows a clear positive relationship between income per person and happiness, where countries with higher income levels generally tend to report higher happiness scores. Since income

data is highly right-skewed, with most countries clustered in the low- to middle-income range and a smaller number of countries having extremely high income levels, a logarithmic scale was applied to the x-axis. After applying the logarithmic scale, the relationship between income and happiness appears more linear and easier to observe. The points follow a clearer upward pattern, indicating that increases in income are associated with increases in happiness. Most low-income countries remain concentrated within the lower happiness range, while middle- and high-income countries are generally located at higher happiness levels. The upward trend line further supports this positive relationship. The R-squared value of 0.68 indicates a strong positive relationship between income and happiness, while the very small p-value (less than 0.0001) suggests that the observed relationship is statistically meaningful and unlikely to occur randomly. Although the overall pattern is consistent, some variation still exists around the trend line, particularly among middle- and high-income countries.

The visualization demonstrates a strong positive relationship between income per person and happiness. In general, countries with higher income levels tend to report higher happiness score

c) Correlation between CPI and Happiness Score



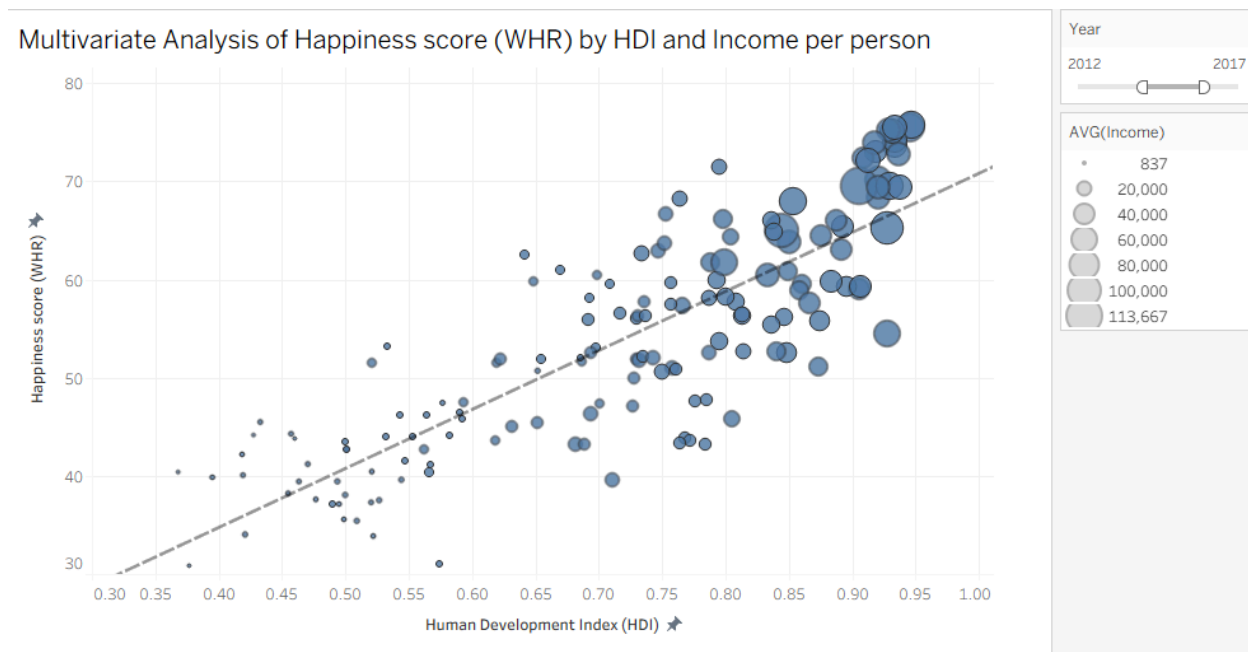
This scatterplot illustrates the relationship between the Corruption Perception Index (CPI) and the Happiness Score (WHR) across countries between 2012 and 2017.

The scatterplot displays a noticeable upward trend, where countries with higher CPI scores generally tend to have higher happiness scores. This pattern may indicate that lower corruption contributes to a more stable social and institutional environment, which can positively influence happiness. Most countries with low CPI values are concentrated in the lower happiness range, while countries with high CPI values are more commonly found in the higher happiness range. Compared to the previous two graphs, the points in this graph are more spread around the trend line, meaning that the relationship between CPI and happiness is less consistent. The graph also suggests that the relationship may become stronger at higher CPI values. Countries with very high CPI scores are more consistently associated with high happiness levels, while countries with low and medium CPI values show greater variation in happiness outcomes. This wider spread indicates that CPI is related

to happiness, but the relationship is not that strong. The R-squared value of 0.5042 indicates a moderate positive relationship between CPI and happiness. In addition, the very small p-value (less than 0.0001) suggests that the relationship shown in the graph is statistically meaningful and unlikely to occur randomly.

Overall, the scatterplot demonstrates a moderate positive relationship between CPI and happiness. Countries with higher CPI scores, representing lower perceived corruption, generally tend to report higher happiness levels.

d) Multivariate Analysis of Happiness Score by HDI and Income



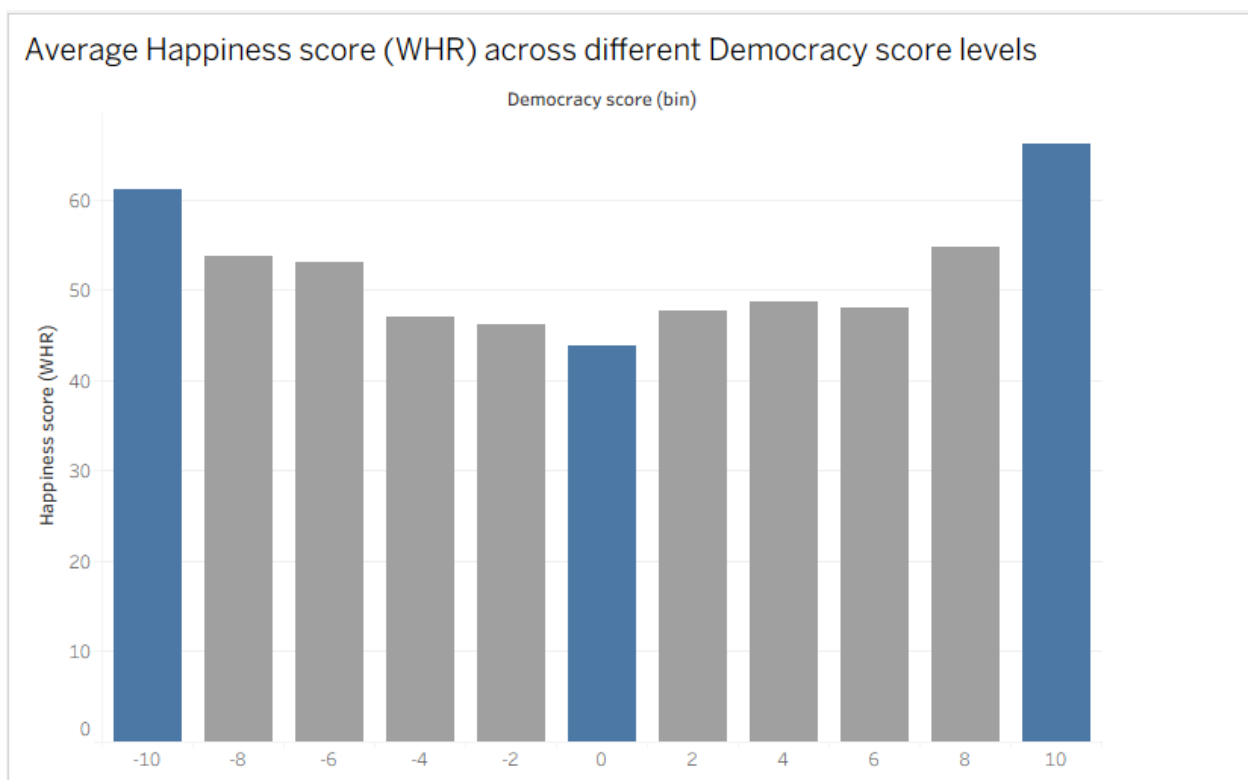
This multivariate scatterplot examines the relationship between the Human Development Index (HDI) and the Happiness Score (WHR) across countries from 2012 to 2017. Bubble sizes represent average income per person.

The graph demonstrates a strong positive relationship between HDI and happiness, with happiness scores generally increasing as HDI values rise. Most countries follow a clear upward pattern, reinforcing the strong association between development and national happiness. In addition, large bubbles, representing countries with higher income per person, are mostly concentrated in the upper-right region of the graph where both HDI and happiness scores are high. In contrast, smaller bubbles are mainly located in the lower-left region, where HDI and happiness scores are lower. This pattern suggests that countries with both strong human development and high income levels tend to achieve the highest happiness scores. The R-squared value of 0.7027 indicates a strong relationship

between HDI and happiness, while the p-value of less than 0.0001 suggests that this relationship is highly reliable and unlikely to occur randomly.

Overall, the graph demonstrates that countries with higher HDI values and higher income per person generally tend to report higher happiness scores.

e. Average Happiness score across different democracy levels



This bar chart displays the average Happiness Score (WHR) across grouped Democracy scores for the period 2005–2011. Democracy scores range from -10, representing the strongest autocracy, to +10, representing the strongest democracy.

Unlike the previous visualizations, this chart does not display a simple linear pattern between democracy and happiness. Countries with the highest democracy score (+10) report the highest average happiness levels. However, countries with the lowest democracy score (-10) also show relatively high happiness scores, exceeding many countries in the middle democracy range. The graph reveals that the lowest happiness levels are concentrated around scores near 0, which represent hybrid or transitional political systems. This may indicate that transitional systems experiencing political instability or economic uncertainty could negatively affect happiness levels. The results suggest that political stability may play an important role in happiness scores, regardless of whether the political system is strongly democratic or strongly authoritarian.

Overall, the visualization suggests that the relationship between democracy and happiness is more complex than the relationships observed with HDI, income, and CPI. While highly democratic countries report the highest average happiness levels, also strongly authoritarian countries also achieve relatively high happiness scores, whereas transitional political systems tend to report lower happiness levels.

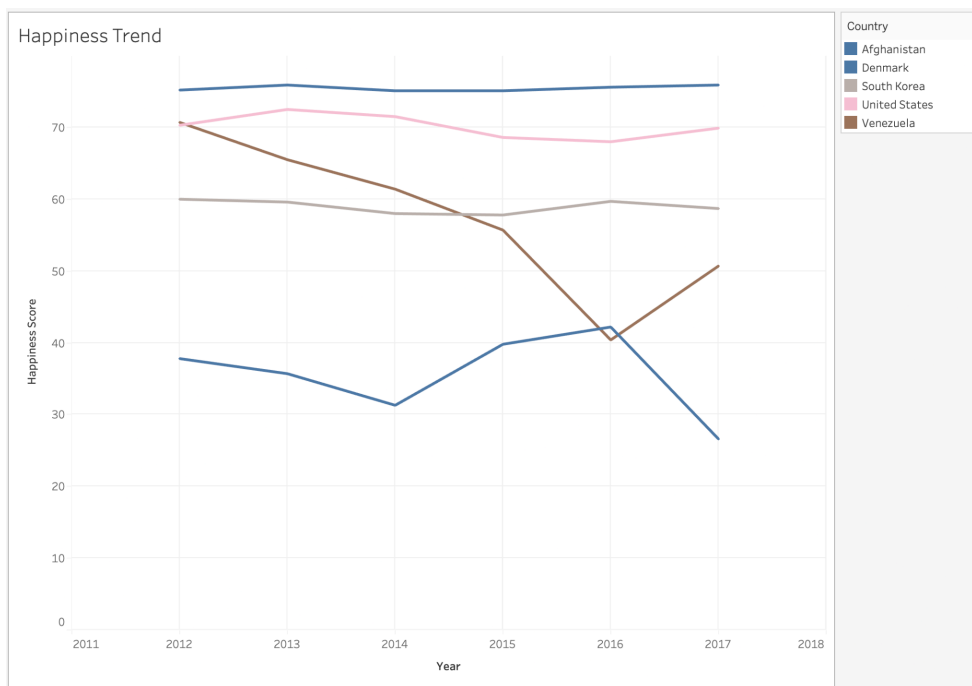
4. Objective 3: Change over Time

Methodology & Scope:

We only considered the time range of 2012-2017 to analyse the link between governance, economic wealth and national well-being. The data density for this particular range is the greatest for each variable. Instead of plotting the entire global picture, that would create a lot of visual clutter, we chose five national archetypes to examine, representing Denmark (high

development/strong democracy), the United States (economic powerhouse/flawed democracy), South Korea (rapid economic developer), Venezuela (institutional crisis), and Afghanistan (low development/conflict baseline).

a) Happiness Score over Time

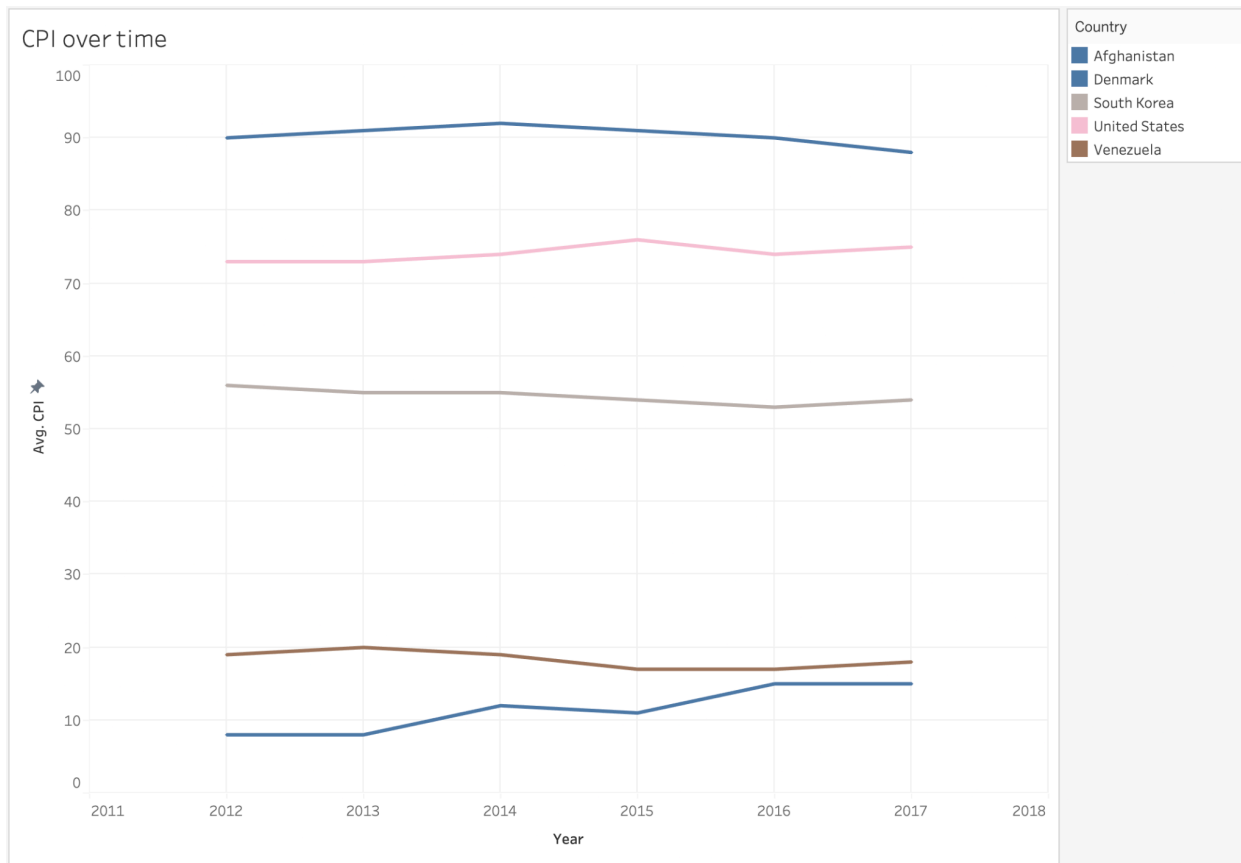


A line chart showing the Average Happiness Score for the past 5 years. Null values for 2014 were filtered to produce a continuous trendline.

Denmark establishes a clear "ceiling" for the dataset, consistently hovering near the top of the global index, while the United States and South Korea maintain relative stability in the middle-upper tier. The most critical narrative emerges from Venezuela. Starting in 2014, Venezuela exhibits a sharp, continuous decline in national happiness. This downward trajectory serves as a lagging indicator of their broader socioeconomic crisis. Afghanistan anchors the bottom of the chart, showing persistently low levels of well-being.

The conclusion is that national happiness is very resilient in a stable and developed democracy, while it is very sensitive when there is a rapid decay of institutions, as Venezuela did.

b) Corruption Perception Index (CPI)

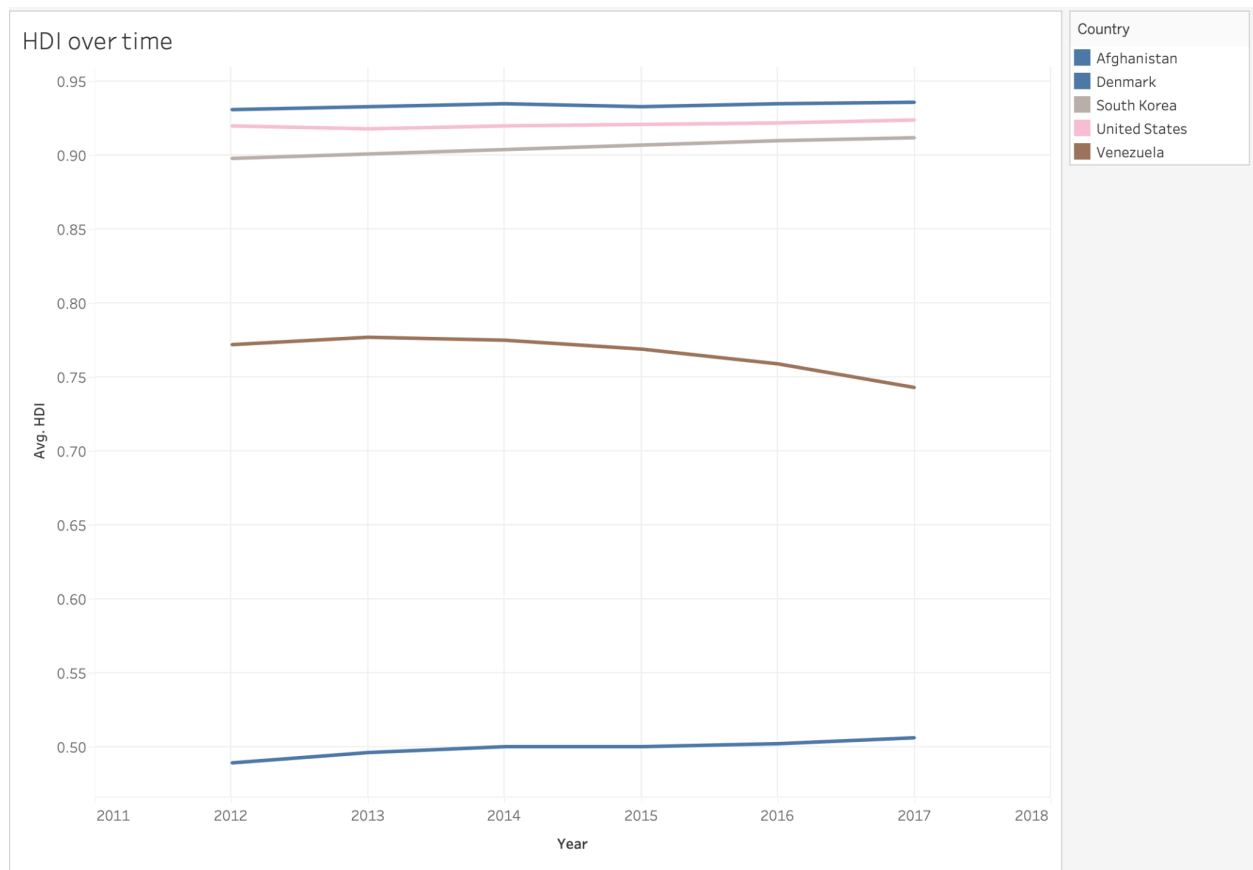


A line chart with the Y axis set to the standard index scale from 0 (highly corrupt) to 100 (very clean) that measures the extent of perceived public sector corruption.

The CPI trends continue to be very consistent across the five-year period, indicating a high level of institutional corruption (or the lack of it) that is very entrenched and difficult to shift. The score for Denmark is always in the 90s (extremely clean), while minor fluctuations can be seen in the score for the USA and South Korea in the 50-75 range. On the other hand, there are two

countries in the bottom of the index that are totally unchanged: Venezuela and Afghanistan (with scores under 20). This corruption correlates strongly with their respective placements at the bottom of the Happiness index.

c) Average HDI over Time



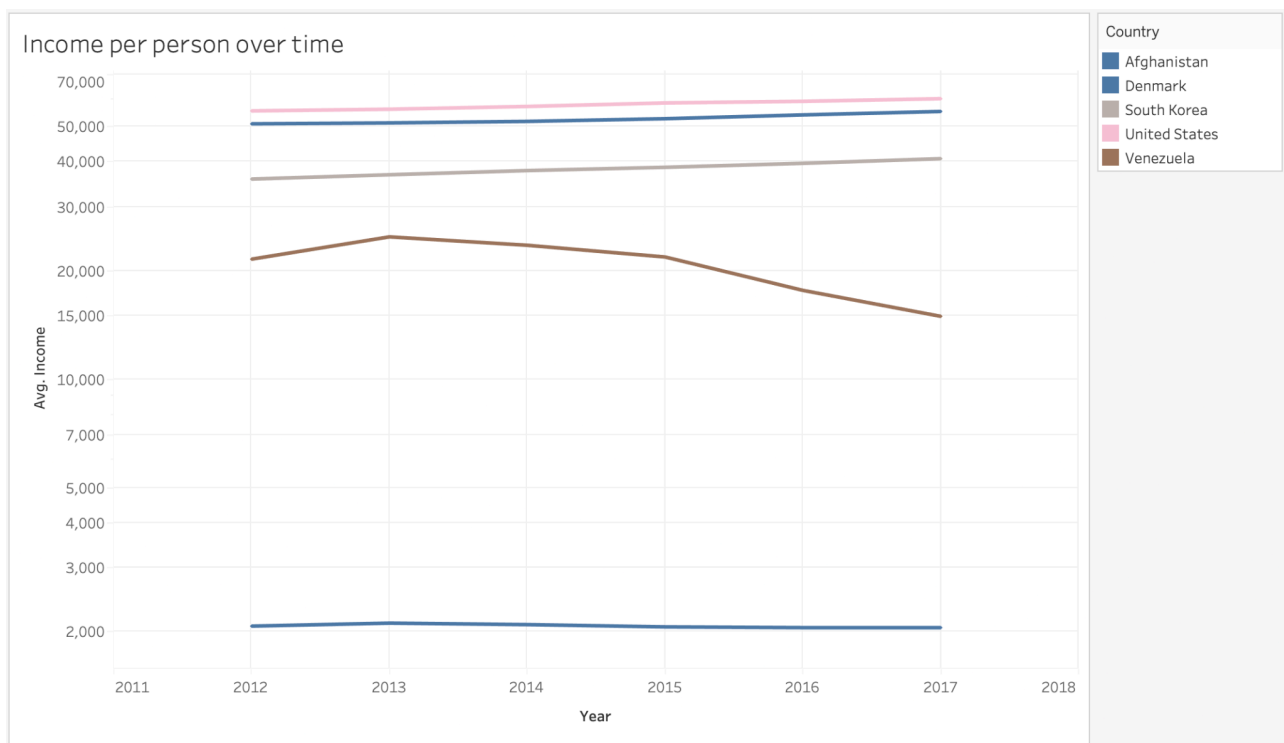
Line chart showing the development of HDI, which is a combination of life expectancy, education and income per person. The Y-axis is auto-scaled (not zero) to emphasize marginal, year-over-year changes.

South Korea is the country that shows the largest rise, indicating its pace of development as a fast-developing country during this time period. Denmark and the US are good examples of slow

but steady growth expected in a well-developed, mature economy. Most importantly, Venezuela's steady growth trajectory falters visibly in the middle of the decade, and serves as a leading indicator of institutional decay before the complete collapse of their economy.

Rapid, compounded improvements in human development are possible in emerging economies such as South Korea, but sustaining that progress is a prerequisite, and the lack of it caused Venezuela's previously stable progress to completely flatline.

d) Average Income over Time

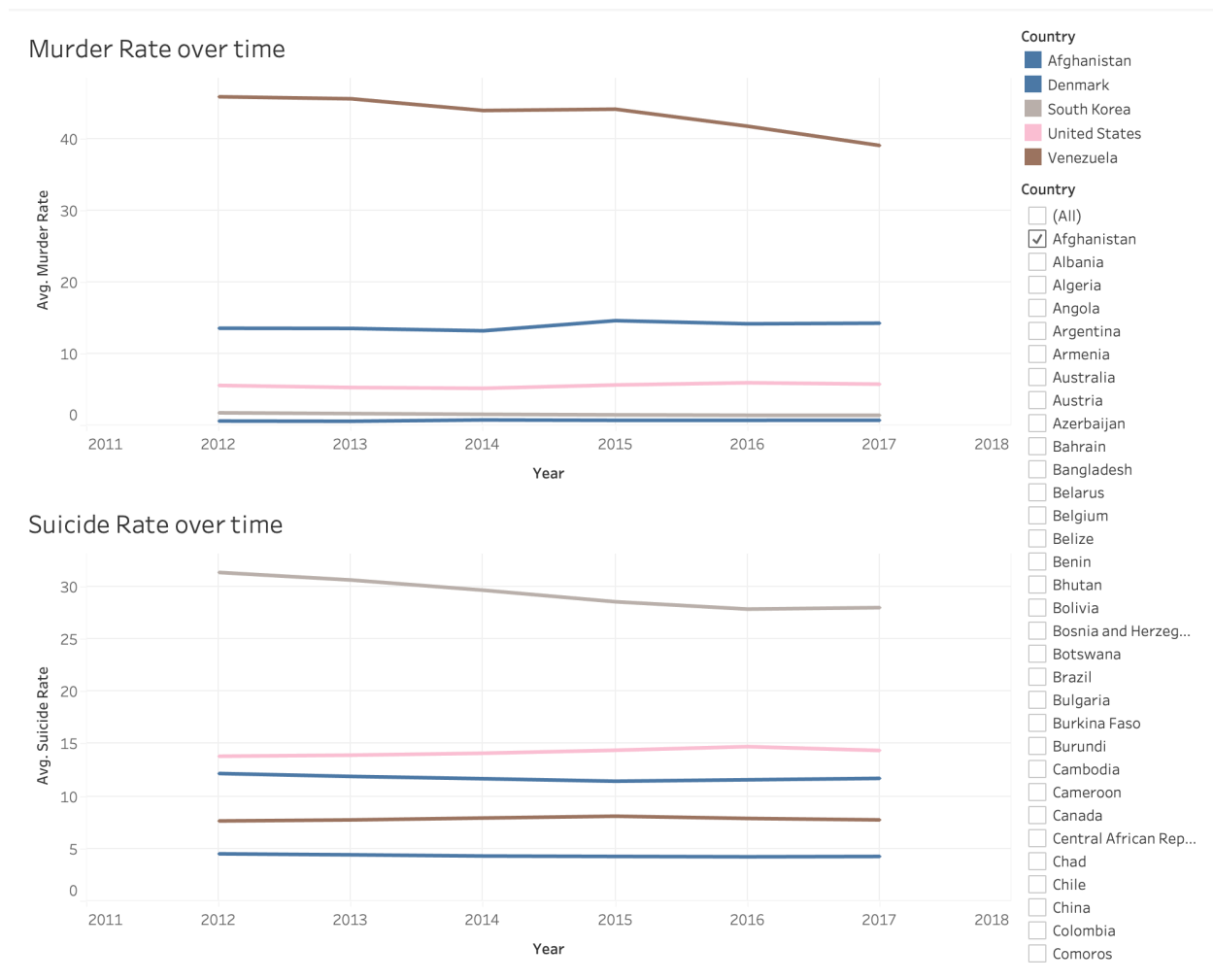


A line chart illustrating GDP per capita (PPP inflation-adjusted). Because of the wide range of global wealth levels, the economic change is shown on a logarithmic scale on the Y axis.

A log-linear graph offers a better view of the economic growth of poorer economies, as it shows the true rate of rise of each economy, rather than flattening out their growth during boom times. The US and Denmark are both high income stable countries, while South Korea indicates

steady growth. What is apparent, however, is Venezuela's dramatic and downward trajectory that began in 2014. This brutal economic downturn is mirrored exactly with their national Happiness score crash at the same time.

e) Murder Vs. Suicide Rate over Time



Line charts side by side, showing the total number of murders and total number of suicides per 100,000 people.

These charts illustrate the flaws in the assumption that wealth equals well-being. The linear scale in the Murder Rate chart enables Venezuela to deviate from the rest of the group. The nation suffers from a homicide explosion just in time for economic and political decline, and this may indicate that the collapse of institutions has a direct physical cost.

On the other hand, the Suicide Rate chart is pointing to the "prosperity paradox. South Korea charts quite a bit apart at the top, trailed by the United States. This indicates that high HDI and Income do not automatically protect a society against the problem of severe mental health problems and societal pressures. This paradox is part of the reason that South Korea's Happiness scores are so low despite its impressive economic and developmental achievements.

These two statistics highlight an important fact about the information: even though increased income and improved governance are very powerful tools for reducing crime and societal breakdown, they are not enough to solve the internal and cultural pressures that lead to extreme mental health issues in well-developed countries.

5.Conclusion

Overall, the project shows that happiness differs greatly across countries and is closely linked to wider national conditions. In objective 1, the data visualizations showed a clear global pattern. The countries with the highest scores were mostly developed, stable, and high income countries, especially in Northern and Western Europe, North America, and Oceania. In contrast, the countries with the lowest happiness scores were more often linked to poverty, conflict, weak institutions, and lower development. The histogram showed that most countries fall in the middle

happiness ranges, while only a smaller group reaches very high happiness levels. The HDI bubble chart also showed that many of the top happiest countries are also among the top HDI countries, meaning health, education, and living standards are strongly connected to national well being. The world map supported this further by showing that happiness is geographically clustered rather than randomly spread.

In objective 2, the relationships visuals gave stronger evidence for the hypothesis. The scatterplot between HDI and happiness showed one of the clearest positive relationships, meaning countries with stronger human development usually reported higher happiness scores. The income and happiness chart also showed a positive relationship, but income alone didn't fully explain happiness because some countries with similar income levels had different happiness scores. The CPI and happiness chart showed that countries with lower perceived corruption generally had higher happiness, suggesting that trust in institutions and government quality matter. The multivariate chart combining HDI, happiness, and income showed that the happiest countries usually had both high development and high income. However, the democracy visualization showed a more complex pattern, meaning democracy alone doesn't explain happiness as HDI, income, or CPI.

In objective 3, the time series visuals showed how happiness and its related factors changed over time. Countries with stable institutions and strong development, such as Denmark and the United States, generally maintained steady happiness levels. South Korea showed how economic and development progress can improve national conditions, but it also revealed that high development doesn't automatically guarantee the highest happiness. Venezuela showed the clearest decline, where falling income, institutional problems, and worsening conditions were reflected in a drop in happiness. Afghanistan remained low across several indicators, showing how long term

conflict and weak development can keep happiness levels low. The murder and suicide rate charts also added important insights, safety and social wellbeing aren't simple. High murder rates can reflect institutional breakdown, while high suicide rates in some developed countries show that economic success doesn't always solve deeper social or mental health pressures.

Based on all three objectives, we reject the null hypothesis because the data shows clear relationships between happiness and several national level factors. The main hypothesis is supported overall, countries with stronger human development, higher income, lower perceived corruption, better governance conditions, and safer social environments generally tend to have higher happiness scores. However, the results also show that happiness isn't caused by one single factor. HDI, income, CPI, murder rate, and suicide rate each explain a part of the picture, but none of them explains happiness completely on its own.

The final insight is that national happiness is a multidimensional outcome. It depends on a combination of economic strengths, human development, institutional trust, political stability, safety, and social well being. Therefore, the project supports the idea that improving happiness requires more than increasing income. Countries also need strong public services, stable institutions, low corruption, safety, and better overall living conditions.